Risk Aversion or Mistaken Beliefs?

Three Martingales and More

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Abstract

This paper explores the confounding of risk aversion and mistaken beliefs in asset pricing. A likelihood ratio—a multiplicative martingale increment—serves as the key mathematical object that connects risk-neutral pricing, subjective belief distortions, and concerns about model misspecification. I describe three uses of likelihood ratios: (i) to encode a representative investor's risk aversion, following Lucas and Hansen; (ii) to represent mistaken beliefs of professional forecasters, following Piazzesi, Salomao, and Schneider; and (iii) to capture fears of model misspecification, following Hansen and Szőke. I show how the third use, with a tilted discounted entropy ball that accommodates concerns about particular parametric models featuring long-run risk, can generate the state-dependent uncertainty prices needed to explain hitherto puzzling term-structure movements.

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1 Introduction

The effects of *risk aversion* and *wrong beliefs* are confounded in asset pricing data. Under rational expectations with a risk-averse representative investor, higher mean returns compensate for higher risks. But if the representative investor holds wrong beliefs, observed average returns depend on *both* risk aversion *and* misunderstood return distributions. Wrong beliefs contribute what look like "stochastic discount factor shocks" and can potentially explain observed countercyclical risk prices.

This paper organizes ideas around a single mathematical device: the *likelihood ratio*, a non-negative random variable with unit mean that twists one probability distribution into another. Likelihood ratios—equivalently, multiplicative martingale increments—appear in at least four distinct roles in modern asset pricing:

Probability	**Likelihood ratio**	**Describes**
Econometric	1	macro risk factors
Risk neutral	m^{λ}_{t+1}	prices of risks
Mistaken	m^{w}_{t+1}	experts' forecasts
Doubtful	$m_{t+1} \in \mathcal{M}$	misspecification fears

Table 1: Four probability measures and associated likelihood ratios. Each likelihood ratio takes the log-normal form $m^{b}_{t+1} = \exp\big(-b_t'\varepsilon_{t+1} - \frac{1}{2}b_t'b_t\big)$ with $b_t = 0$, λ_t, w_t, or a worst-case distortion.

The rest of the paper proceeds as follows. Section 2 sets up the Gaussian framework in which a likelihood ratio twists probabilities while preserving a tractable structure. Section 3 reviews rational-expectations asset pricing and shows how a single likelihood ratio turns risk-neutral pricing formulas into modern asset pricing formulas. Section 4 discusses the identification challenge: the same likelihood ratio can be interpreted as risk aversion or as belief distortion. Section 5 describes the approach of Piazzesi, Salomao, and Schneider (2015), who use survey data on professional forecasters to decompose the likelihood ratio into a smaller risk price and a belief distortion. Section 7 develops the robust control theory approach, beginning with Hansen's formulation with a discounted entropy ball and culminating in Hansen and Szőke's tilted entropy ball that accommodates concerns about parametric alternatives with more long-run risk. Section 9 summarizes empirical results. Section 11 concludes with reflections on why these joint probability distributions matter for macroeconomics.

2 Risks and Beliefs

Let ε denote a vector of risks to be taken and priced. Suppose that, under the econometrician's probability model, ε has a standard multivariate normal density:

$$\phi(\varepsilon) \propto \exp\left(-\tfrac{1}{2}\,\varepsilon'\varepsilon\right), \qquad \varepsilon \sim \mathcal{N}(0, I).$$

Define a *likelihood ratio*

$$m(\varepsilon) = \exp\left(-\lambda'\varepsilon - \tfrac{1}{2}\,\lambda'\lambda\right) \geq 0,$$

which satisfies $E\,m(\varepsilon) = 1$ by construction. The *twisted density*

$$\hat{\phi}(\varepsilon) = m(\varepsilon)\,\phi(\varepsilon) \propto \exp\left(-\tfrac{1}{2}\,(\varepsilon+\lambda)'(\varepsilon+\lambda)\right)$$

is a $\mathcal{N}(-\lambda, I)$ density. The *relative entropy* of the twisted density with respect to the baseline density is

$$E\big[m(\varepsilon)\log m(\varepsilon)\big] = \tfrac{1}{2}\,\lambda'\lambda,$$

a convenient measure of the statistical distance between the two models.

The vector λ is the key object. Depending on context it represents risk prices, belief distortions, or worst-case mean perturbations under model uncertainty.

3 Asset Pricing with Likelihood Ratios

3.1 The econometrician's model

The econometrician works with a linear Gaussian state-space system:

$$x_{t+1} = A\,x_t + C\,\varepsilon_{t+1}, \tag{1}$$

$$y_{t+1} = D\,x_t + G\,\varepsilon_{t+1}, \tag{2}$$

$$\varepsilon_{t+1} \sim \mathcal{N}(0, I),$$

where y_{t+1} collects utility-relevant variables, $r_t = \bar{r}\,x_t$ is the risk-free one-period interest rate, and $d_t = \bar{d}\,x_t$ is the payout process from an asset. The object of interest is the price at date t of a claim to the random payout stream $\{d_{t+j}\}_{j=1}^{\infty}$.

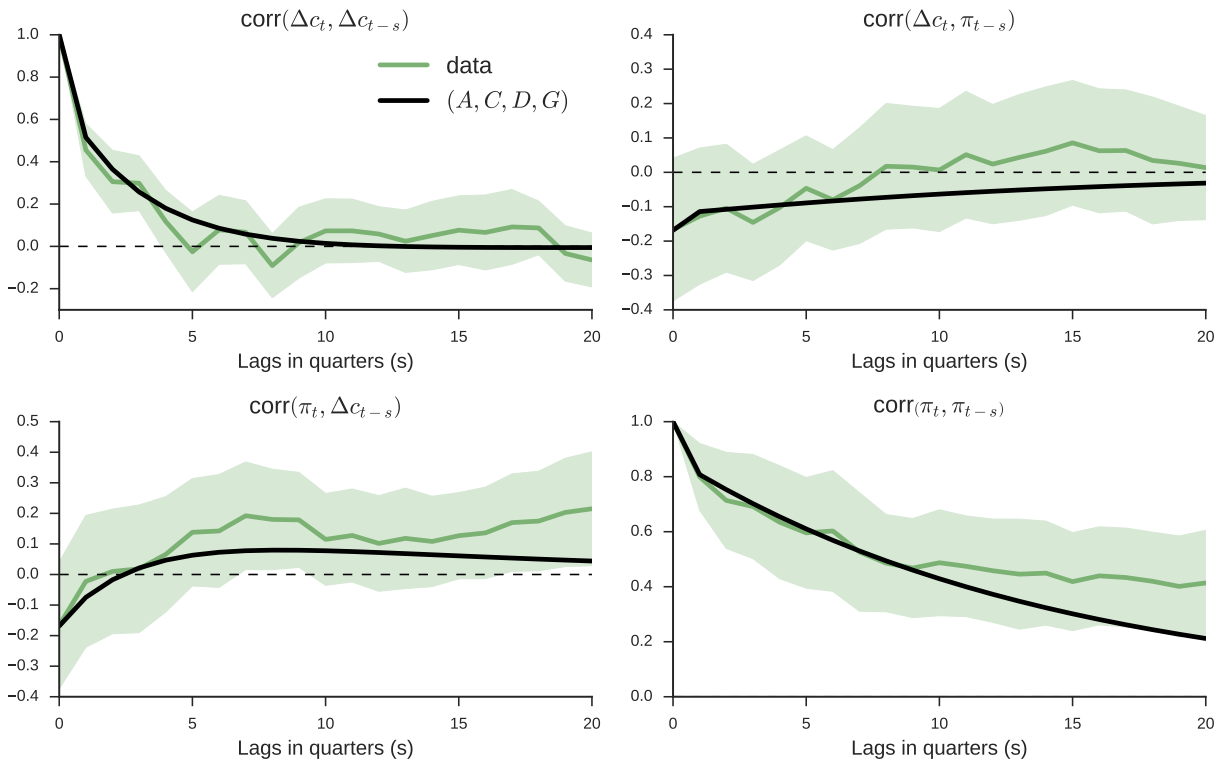

Figure 1: The econometrician's model: estimated state dynamics.

3.2 Risk-neutral rational expectations pricing

Under rational expectations with a risk-neutral representative investor, stock prices satisfy (Shiller):

$$p_t = \exp(-r_t)\, E_t\big(p_{t+1} + d_{t+1}\big).$$

The expectations theory of the term structure of interest rates (Hicks–Shiller) prices a zero-coupon risk-free claim to one dollar at time $t + n$ as:

$$\begin{aligned} p_t(1) &= \exp(-r_t), \\ p_t(n+1) &= \exp(-r_t)\, E_t\, p_{t+1}(n), \\ p_t(n) &= \exp(B_n\, x_t). \end{aligned} \tag{3}$$

These formulas work "pretty well" for conditional means but less well for conditional variances—the Shiller "volatility puzzles."

3.3 Modern asset pricing: adding risk aversion

It would be convenient if versions of the same formulas worked even when investors are risk averse or do not have rational expectations. The likelihood ratio makes this possible.

Let

$$\begin{aligned} m^{\lambda}_{t+1} &= \exp\bigl(-\lambda_t' \, \varepsilon_{t+1} - \tfrac{1}{2}\, \lambda_t' \lambda_t\bigr)\,, \qquad \lambda_t = \lambda \, x_t, \\ E_t \, m^{\lambda}_{t+1} &= 1, \qquad m^{\lambda}_{t+1} \geq 0. \end{aligned} \tag{4}$$

The likelihood ratio m^{λ}_{t+1} distorts the conditional distribution of ε_{t+1} from $\mathcal{N}(0, I)$ to $\mathcal{N}(-\lambda \, x_t, I)$. Covariances of returns with m^{λ}_{t+1} affect mean returns—this is the channel through which risk aversion prices risks.

With this device, modern asset pricing (Lucas–Hansen for stocks, Dai–Singleton–Backus–Zin for bonds) takes the form:

$$p_t = \exp(-r_t) \, E_t\bigl(m^{\lambda}_{t+1} \, (p_{t+1} + d_{t+1})\bigr), \tag{5}$$

and for the term structure:

$$\begin{aligned} p_t(1) &= \exp(-r_t), \\ p_t(n+1) &= \exp(-r_t) \, E_t\bigl(m^{\lambda}_{t+1} \, p_{t+1}(n)\bigr), \\ p_t(n) &= \exp(B_n \, x_t). \end{aligned} \tag{6}$$

3.4 Risk-neutral dynamics

The risk-neutral representation implies twisted dynamics:

$$x_{t+1} = (A - C\lambda) \, x_t + C \, \tilde{\varepsilon}_{t+1}, \qquad \tilde{\varepsilon}_{t+1} \sim \mathcal{N}(0, I).$$

The risk-neutral dynamics assert that the shock distribution ε_{t+1} has conditional mean $-\lambda_t$ instead of 0. The dependence of $\lambda_t = \lambda \, x_t$ on the state modifies the dynamics relative to the econometrician's model.

3.5 Expectation under a twisted distribution

The mathematical expectation of y_{t+1} under the probability distribution twisted by likelihood ratio m_{t+1} is

$$\tilde{E}_t \, y_{t+1} = E_t \, m_{t+1} \, y_{t+1}.$$

With respect to the risk-neutral dynamics, the term structure theory becomes:

$$
\begin{aligned}
p_t(1) &= \exp(-r_t), \\
p_t(n+1) &= \exp(-r_t)\, \tilde{E}_t\, p_{t+1}(n), \\
p_t(n) &= \exp(\tilde{B}_n\, x_t).
\end{aligned}
$$

These are the same formulas as rational-expectations asset pricing, but mathematical expectations are taken with respect to a probability measure twisted by risk aversion.

4 The Identification Challenge

The vector λ_t can be interpreted as either:

- a risk price vector expressing the representative agent's risk aversion, or
- the representative agent's belief distortion relative to the econometrician's model.

The asset pricing formulas (5)–(6) are identical under both interpretations, and so are the econometric fits. Relative to the model of a risk-averse representative investor with rational expectations, a model of a risk-neutral investor with appropriately mistaken beliefs produces observationally equivalent predictions. This insight was articulated by Hansen, Sargent, and Tallarini (1999) and Piazzesi, Salomao, and Schneider (2015).

To distinguish risk aversion from belief distortion, one needs either more information (the PSS approach using survey data) or more theory (the Hansen and Szőke robust control approach), or both (Szőke).

5 More Information: Experts' Forecasts

5.1 The PSS framework

Piazzesi, Salomao, and Schneider (2015, henceforth PSS) exploit data on professional forecasters' expectations to decompose the likelihood ratio into risk prices and belief distortions. Their setup posits:

- The representative agent's risk aversion leads him to price risks ε_{t+1} with prices $\lambda_t^* = \lambda^*\, x_t$.
- The representative agent has twisted beliefs $(A^*, C) = (A - Cw^*, C)$ relative to the econometrician's model (A, C).

- Professional forecasters use the twisted beliefs (A^*, C) to answer survey questions about their forecasts.

The question PSS assume experts think they are answering is: "What forecasts underlie your investment advice decisions?"

5.2 Estimation strategy

PSS proceed as follows:

1. Use data $\{x_t\}_{t=0}^T$ to estimate the econometrician's model A, C.
2. Project experts' forecasts $\{\hat{x}_{t+1}\}$ on x_t to obtain $\hat{x}_{t+1} = A^* x_t$ and interpret A^* as a belief distortion.
3. Back out the mean distortion $w^* x_t = -C^{-1}(A^* - A)\, x_t$ to the density of ε_{t+1}.
4. Reinterpret the λ estimated by the rational-expectations econometrician as $\lambda = \lambda^* + w^*$, where $\lambda_t^* = \lambda^* x_t$ is the (smaller) price of risk vector actually charged by the representative agent with distorted beliefs.

An econometrician who mistakenly imposes rational expectations estimates risk prices λ_t that sum two parts:

- smaller risk prices λ_t^* actually charged by the erroneous-beliefs representative agent, and
- conditional mean distortions w_t^* of the risks ε_{t+1} that the twisted-beliefs representative agent's model displays relative to the econometrician's.

5.3 Empirical specification

The key variables in PSS's state-space system are:

- the level and slope of the yield curve (short rate; spread between 5-year and short rate),
- inflation, and
- conditional expectations of these variables.

5.4 PSS successes

PSS find $w^* \neq 0$:

- The expert's model differs systematically from the econometrician's.
- Experts perceive the level and slope of the yield curve to be more persistent than the econometrician's estimates imply.
- Subjective risk prices $\lambda^* x_t$ vary less than the λx_t estimated by the rational-expectations econometrician.

5.5 Why are beliefs distorted?

PSS offer no explanation for why beliefs are distorted. Are they mistakes? Ignorance of good econometrics? And how distorted are they in terms of the divergence of probabilities? These questions motivate a theory of belief distortions rooted in robust control theory.

6 Rational Expectations and Its Limits

6.1 The 1970s–1980s informal justification

The standard justification for rational expectations treats it as the outcome of learning from an infinite history: least-squares learning converges to rational expectations. The argument requires that:

- agents know correct functional forms, and
- a stochastic approximation argument partitions dynamics into fast parts (justifying a law of large numbers) and slow parts (justifying an ODE).

However, long intertemporal dependencies make rates of convergence slow.

6.2 Good econometricians

Good econometricians have limited data and only hunches about functional forms. They fear that their fitted models are incorrect. An agent who is like a good econometrician:

- has a parametric model estimated from limited data,

- acknowledges that many other specifications fit nearly as well—other parameter values, other functional forms, omitted variables, neglected nonlinearities, and history dependencies,
- fears that one of those other models actually prevails, and
- seeks "good enough" decisions under *all* such alternative models—*robustness*.

7 A Theory of Belief Distortions: Robust Control

7.1 Hansen's dubious agent

Inspired by robust control theory, consider a dubious investor who:

- shares the econometrician's model A, C, D, G,
- expresses doubts by using a continuum of likelihood ratios to form a *discounted entropy ball* of size η around the econometrician's model,
- wants a valuation that is good for every model in the entropy ball, and
- constructs a lower bound on values and a worst-case model that attains it.

7.1.1 Why discounted entropy?

Discounted entropy includes models that undiscounted entropy excludes. Undiscounted entropy over infinite sequences excludes many models that are very difficult to distinguish from the econometrician's model with limited data, because undiscounted entropy includes only models that share laws of large numbers.

7.2 Valuation under the econometrician's model

Taking the log consumption process to be linear Gaussian with shocks $\varepsilon_{t+1} \sim \mathcal{N}(0, I)$:

$$c_{t+1} - c_t = D\, x_t + G\, \varepsilon_{t+1}, \tag{7}$$

$$x_{t+1} = A\, x_t + C\, \varepsilon_{t+1}, \tag{8}$$

the value function is

$$V(x_0, c_0) := E\left[\sum_{t=0}^{\infty} \beta^t\, c_t \;\middle|\; x_0, c_0\right] = c_0 + \beta\, E[V(x_1, c_1) \mid x_0, c_0]\,.$$

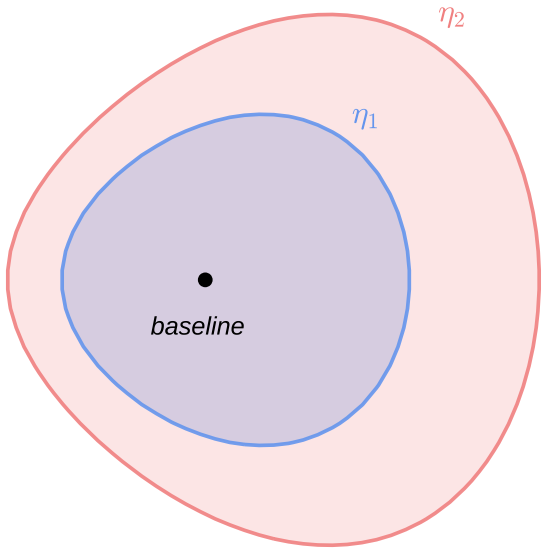

Figure 2: Discounted entropy ball around the econometrician's model.

7.3 The sequence problem

The dubious agent solves:

$$
\begin{aligned}
J(x_0, c_0 \mid \eta) := \min_{\{m_{t+1}\}_{t=0}^{\infty}} \; & E\left[\sum_{t=0}^{\infty} \beta^t \, M_t \, c_t \;\middle|\; x_0, c_0\right] \\
\text{s.t.} \quad & c_{t+1} - c_t = D\, x_t + G\, \varepsilon_{t+1}, \\
& x_{t+1} = A\, x_t + C\, \varepsilon_{t+1}, \\
& E\left[\sum_{t=0}^{\infty} \beta^t \, M_t \, E[m_{t+1} \log m_{t+1} \mid x_t, c_t] \;\middle|\; x_0, c_0\right] \leq \eta, \\
& M_{t+1} = M_t \, m_{t+1}, \quad E[m_{t+1} \mid x_t, c_t] = 1, \quad M_0 = 1.
\end{aligned}
\tag{9}
$$

The likelihood ratio process $\{M_t\}_{t=0}^{\infty}$ is a multiplicative martingale.

7.4 Entropy and the likelihood ratio

With the log-normal likelihood ratio

$$m_{t+1} := \exp\left(-\tfrac{w_t' w_t}{2} - w_t' \, \varepsilon_{t+1}\right),$$

conditional entropy takes the simple form

$$E[m_{t+1} \log m_{t+1} \mid x_t, c_t] = \tfrac{1}{2} \, w_t' w_t.$$

Substituting this into (9) yields the reformulated problem:

$$\begin{aligned} J(x_0, c_0 \mid \eta) := \min_{\{w_t\}_{t \geq 1}} \; & E^w \left[\sum_{t=0}^{\infty} \beta^t \, c_t \;\middle|\; x_0, c_0 \right] \\ \text{s.t.} \quad & c_{t+1} - c_t = D \, x_t + G \, (\tilde{\varepsilon}_{t+1} - w_t), \qquad \tilde{\varepsilon}_{t+1} \sim \mathcal{N}(0, I), \\ & x_{t+1} = A \, x_t + C \, (\tilde{\varepsilon}_{t+1} - w_t), \\ & \tfrac{1}{2} \, E^w \left[\sum_{t=0}^{\infty} \beta^t \, w_t' w_t \;\middle|\; x_0, c_0 \right] \leq \eta. \end{aligned} \tag{10}$$

7.5 Outcome of the Hansen model

The worst-case mean distortion turns out to be a constant vector:

$$w_t = \bar{w}.$$

The consequence is that the contribution of w_t to risk prices is state-independent. This does *not* help explain countercyclical prices of risk (or prices of model uncertainty).

8 Tilting the Entropy Ball

8.1 Hansen and Szőke's more refined dubious agent

To generate state-dependent uncertainty prices, Hansen and Szőke introduce a more refined dubious agent who:

- shares the econometrician's model A, C, D, G,
- expresses doubts by using a continuum of likelihood ratios to form a discounted entropy ball around the econometrician's model,

- also insists that some martingales representing particular alternative *parametric* models be included in the discounted entropy ball.

The inclusion of those alternative parametric models *tilts* the entropy ball, which affects the worst-case model in a way that can produce countercyclical uncertainty prices.

8.2 Concern about other parametric models

The investor wants to include particular alternative models with

$$E_t[\bar{m}_{t+1} \log \bar{m}_{t+1}] = \tfrac{1}{2}\, \bar{w}_t' \, \bar{w}_t = \xi(x_t)$$

and discounted entropy

$$E^{\bar{w}} \left[\sum_{t=0}^{\infty} \beta^t \, \xi(x_t) \;\middle|\; x_0, c_0 \right].$$

This is accomplished by replacing the earlier entropy constraint with

$$\tfrac{1}{2}\, E^{w} \left[\sum_{t=0}^{\infty} \beta^t \, w_t' w_t \;\middle|\; x_0, c_0 \right] \leq E^{w} \left[\sum_{t=0}^{\infty} \beta^t \, \xi(x_t) \;\middle|\; x_0, c_0 \right]. \tag{11}$$

8.3 Yaron's bazooka

Bansal and Yaron (2004) mix long-run risk (LRR) with Epstein–Zin (EZ) preferences. LRR is statistically difficult to detect and estimate, but an Epstein–Zin agent or a dubious agent really hates LRR. Under certain conditions, the EZ value function is the indirect utility function of a dubious agent. This sets the stage for big risk prices—but not countercyclical ones.

8.4 Concern about bigger long-run risk in the Szőke model

Inspired by Bansal and Yaron (2004), an agent fears particular long-run risks expressed by

$$x_{t+1} = \bar{A}\, x_t + C\, \tilde{\varepsilon}_{t+1}.$$

This corresponds to $\bar{w}_t = \bar{w}\, x_t$ with

$$\bar{w} = -C^{-1}(\bar{A} - A),$$

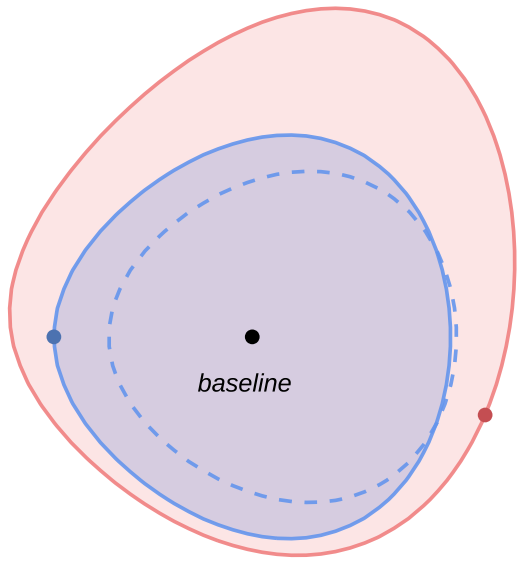

Figure 3: Tilted discounted entropy balls. Including particular parametric alternatives with more long-run risk tilts the entropy ball and generates state-dependent worst-case distortions.

which implies a quadratic ξ function:

$$\xi(x_t) := x_t' \, \bar{w}' \, \bar{w} \, x_t =: x_t' \, \Xi \, x_t.$$

8.5 State-dependent contributions to the entropy constraint

The time-t contributions to the right-hand side of (11) relax the discounted entropy constraint in states x_t in which $\xi(x_t)$ is larger. This sets the stage for state-dependent mean distortions in the worst-case model, which in turn ignite countercyclical market prices of uncertainty.

8.6 The Szőke agent's sequence problem

The resulting linear-quadratic problem is:

$$J(x_0, c_0 \mid \Xi) := \max_{\tilde{\theta} \geq 0} \min_{\{w_t\}_{t \geq 1}} E^w \left[\sum_{t=0}^{\infty} \beta^t c_t + \tilde{\theta} \, \tfrac{1}{2} \sum_{t=0}^{\infty} \beta^t \big(w_t' w_t - x_t' \, \Xi \, x_t \big) \;\middle|\; x_0, c_0 \right] \tag{12}$$

$$\begin{aligned} \text{s.t.} \quad & c_{t+1} - c_t = D \, x_t + G \, (\tilde{\varepsilon}_{t+1} - w_t), \qquad \tilde{\varepsilon}_{t+1} \sim \mathcal{N}(0, I), \\ & x_{t+1} = A \, x_t + C \, (\tilde{\varepsilon}_{t+1} - w_t). \end{aligned}$$

The worst-case shock mean distortion is

$$\tilde{w}_t = \tilde{w} \, x_t,$$

and the worst-case model is $(\tilde{A}, C, \tilde{D}, G)$ with

$$\begin{aligned} \tilde{A} &= A - C \, \tilde{w}, \\ \tilde{D} &= D - G \, \tilde{w}. \end{aligned}$$

9 Empirical Results

9.1 The U.S. term structure

Several recognized patterns characterize the U.S. term structure:

- The nominal yield curve usually slopes upward.
- The long-minus-short yield spread narrows before U.S. recessions and widens after them.
- Consequently, the slope of the yield curve helps predict aggregate inputs and outputs.
- Long and short yields are almost equally volatile (the "Shiller puzzle").
- To solve the Shiller puzzle, risk prices (or something observationally equivalent) must depend on volatile state variables.

9.2 Challenges and responses

Table 2 summarizes how various models perform against these empirical challenges.

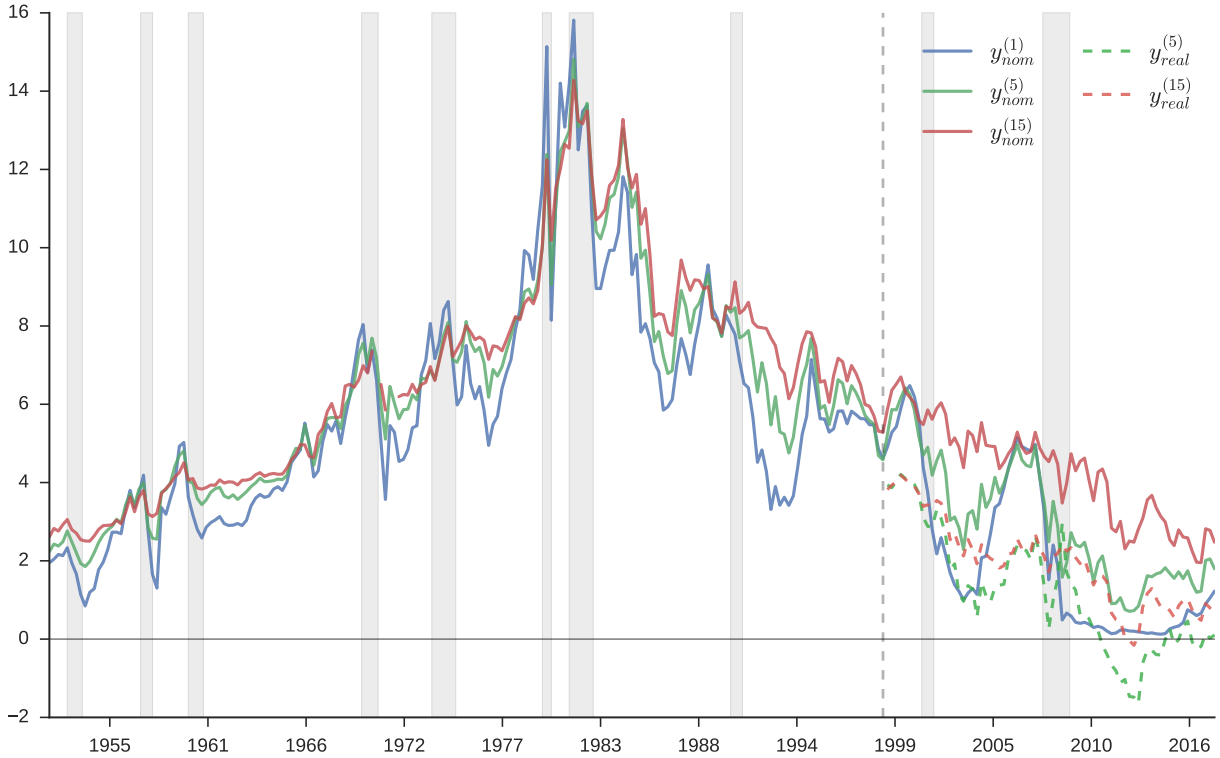

Figure 4: U.S. term structure of interest rates.

	Average slope	Slopes near recessions	Volatile long yield
Lucas (1978)	no	no	no
Epstein–Zin with LRR (PS (2007), HS (2001))	maybe	yes	no
PSS (2015)	built-in	built-in	yes
Szőke (2017)	YES	yes	yes

Table 2: Challenges and model performances. Affine risk prices with persistent consumption growth can address the second column; the third column requires state-dependent prices of risk.

9.3 Szőke's empirical strategy

Szőke proceeds in two stages:

Stage I: Estimation.

1. Use $\{x_t, c_t\}_{t=0}^T$ to estimate the econometrician's A, C, D, G.

2. View Ξ as a matrix of additional free parameters and estimate them simultaneously with risk prices $\tilde{\lambda}\, x_t$ from data $\{p_t(n+1)\}_{n=1}^N$, $t = 0, \ldots, T$, by imposing cross-equation restrictions:

$$
\begin{aligned}
p_t(n+1) &= \exp(-r_t)\, E_t\Big[m_{t+1}^{\tilde{w}}\, m_{t+1}^{\tilde{\lambda}}\, p_{t+1}(n)\Big]\,, \\
m_{t+1}^{\tilde{w}} &= \exp\Big(-\tilde{w}_t'\, \varepsilon_{t+1} - \tfrac{\tilde{w}_t'\, \tilde{w}_t}{2}\Big)\,, \\
m_{t+1}^{\tilde{\lambda}} &= \exp\Big(-\tilde{\lambda}_t'\, \varepsilon_{t+1} - \tfrac{\tilde{\lambda}_t'\, \tilde{\lambda}_t}{2}\Big)\,,
\end{aligned}
\tag{13}
$$

   where E_t is taken with respect to the econometrician's model and $\tilde{w}_t = \tilde{w}\, x_t$ is the dubious investor's worst-case model.

Stage II: Assessment.

1. Assess improvements in predicted behavior of the term structure.

2. Use estimated worst-case dynamics to form distorted forecasts $\tilde{x}_{t+1} = (A - C\,\tilde{w})\, x_t$ and compare them to those of professional forecasters.

3. Compute discounted relative entropy of the worst-case twisted model $(A - C\,\tilde{w}, C, D - G\,\tilde{w}, G)$ relative to the econometrician's model (A, C, D, G), and use it together with Chernoff–Newman–Stuck entropy measures to assess the difficulty of distinguishing the two models.

9.4 Interpretations of conditional mean distortions

The conditional mean distortions $w\, x_t$ admit distinct interpretations:

- In PSS, w_t^* is a vector of "mistakes" or "suboptimal forecasts."

- In Szőke, $\tilde{w}_t$ is a vector of "model uncertainty" prices: compensations the representative agent charges to bear risks ε_{t+1} with unknown probability distribution.

9.5 Insights from Szőke's model

Szőke's framework delivers:

1. A theory of state-dependent belief distortions $\tilde{w}_t = \tilde{w}\, x_t$.

2. A theory about the question that professional forecasters answer: they respond with their worst-case model because they hear "tell me forecasts that rationalize your (max-min) decisions."

3. A way to measure the size of belief distortions relative to the econometrician's model.

He uses his estimated Ξ matrix:

- to infer an equivalence class of alternative parametric models parameterized by $\bar{w}$ that concern the representative investor—these models have more long-run risk than the econometrician's model;

- to infer a worst-case mean distortion $\tilde{w}\, x_t$ whose state dependence causes the term structure to move with x_t in ways that explain hitherto puzzling term-structure movements, including Shiller's "volatility puzzle."

9.6 Three probability twisters

To summarize, three distinct probability twisters play roles in this analysis:

- w_t^*: Piazzesi, Salomao, and Schneider's mistaken agent;

- $\bar{w}_t$: Szőke's especial LRR parametric worry;

- $\tilde{w}_t$: Szőke's worst-case model.

10 Cross-Equation Restrictions and Estimation

A key appeal of the robust control approach is that it lets us deviate from rational expectations while still preserving a set of powerful cross-equation restrictions on decision makers' beliefs. As Szőke puts it:

> An appealing feature of robust control theory is that it lets us deviate from rational expectations, but still preserves a set of powerful cross-equation restrictions on decision makers' beliefs... Consequently, estimation can proceed essentially as with rational expectations econometrics. The main difference is that now restrictions through which we interpret the data emanate from the decision maker's best response to a worst-case model instead of to the econometrician's model.

11 Who Cares?

Joint probability distributions of interest rates and macroeconomic shocks are important throughout macroeconomics:

- **Costs of aggregate fluctuations.** Welfare assessments of business cycles depend sensitively on how risks are priced.
- **Consumption Euler equations.** The "New Keynesian IS curve" is a log-linearized consumption Euler equation whose risk adjustments are controlled by the stochastic discount factor.
- **Optimal taxation and government debt management.** Government bond prices embed risk prices whose state dependence matters for optimal fiscal policy.
- **Central bank expectations management.** Forward guidance works by shifting the term structure, an exercise whose effects depend on the same likelihood ratios studied here.
- **Long-run risk and secular stagnation.** The Bansal–Yaron long-run risk hypothesis is difficult to detect statistically, yet an agent who fears it in the sense formalized above may behave very differently than one who does not.

Understanding whether observed asset prices reflect risk aversion, mistaken beliefs, or fears of model misspecification—and quantifying each component—is therefore essential for both positive and normative macroeconomics.

A Multiplier Preferences

The multiplier preference version of the dubious agent's problem is:

$$W(x_0, c_0 \mid \theta) := \min_{\{m_{t+1}\}_{t\geq 0}} E\left[\sum_{t=0}^{\infty} \beta^t M_t\big(c_t + \theta\, m_{t+1} \log m_{t+1}\big) \,\middle|\, x_0, c_0\right], \tag{14}$$

$$M_{t+1} = M_t\, m_{t+1}, \quad E[m_{t+1} \mid x_t, c_t] = 1, \quad M_0 = 1.$$

The recursive formulation is:

$$
\begin{aligned}
W(x_t, c_t \mid \theta) &= c_t + \min_{m_{t+1}} \; E\big[m_{t+1}\big[\beta W(x_{t+1}, c_{t+1}) + \theta \log m_{t+1}\big] \;\big|\; x_t, c_t\big] \\
&= c_t - \theta \log E\left[\exp\left(-\frac{\beta W(x_{t+1}, c_{t+1})}{\theta}\right) \;\middle|\; x_t, c_t\right] \\
&=: c_t + T_t[\beta W(x_{t+1}, c_{t+1})]\,, \qquad (15)
\end{aligned}
$$

where the right-hand side is attained by

$$
m^*_{t+1} \propto \exp\left(-\frac{W(x_{t+1}, c_{t+1})}{\theta}\right).
$$

Relationship between multiplier and constraint problems. By Lagrange multiplier theory,

$$
W(x_t, c_t \mid \tilde{\theta}) = J(x_t, c_t \mid \eta) + \tilde{\theta}\,\eta.
$$